

Week 5 - Task 13: Phishing Email Analysis

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1. Objective

Analyze a phishing email as a SOC L1 analyst would: inspect headers, verify sender authenticity, check a suspicious URL via VirusTotal (without clicking it), list red flags, and issue a final verdict.

2. Header Analysis

Field	Observation
From	Display name did not match the actual sender domain
Return-Path	Different domain than the "From" address (mismatch)
SPF	Fail — sending server not authorized for the claimed domain
DKIM	Fail / missing signature — email not cryptographically verified

Finding: SPF/DKIM misalignment is a strong spoofing indicator. Legitimate organizations authenticate their mail; this message did not pass authentication checks.

3. URL Investigation (VirusTotal)

The suspicious link was submitted to VirusTotal for passive analysis (not clicked directly).

- URL checked: `http://example.com/login`
- Detection result: **0/90 vendors flagged it malicious**
- Status: 404, Content-Type: text/html
- Crowdsourced context: 1 low-severity historical note (unrelated old threat report), otherwise rated benign/legitimate by contextual indicators

See `thm3.png` and `thm4.png` for the full vendor breakdown (all "Clean"/"Unrated").

Finding: This specific placeholder URL returned clean. In a real phishing case, a malicious redirect chain, mismatched domain, or IP-based link would show up here — this is the correct workflow to use in real triage.

4. Red Flags Identified

- Sender domain does not match the organization it claims to represent
- SPF and DKIM authentication failed
- Urgency/pressure language typical of phishing pretexts
- Link domain inconsistent with the claimed sender

5. Verdict

Phishing indicators present at the header/authentication level (SPF/DKIM fail + domain mismatch). The specific URL tested came back clean on VirusTotal, showing the correct method to verify a link safely — in a live scenario, a failed SPF/DKIM combined with a malicious URL would confirm phishing; here, the header analysis alone is sufficient to flag the email as suspicious and hold it for further review rather than trust it.

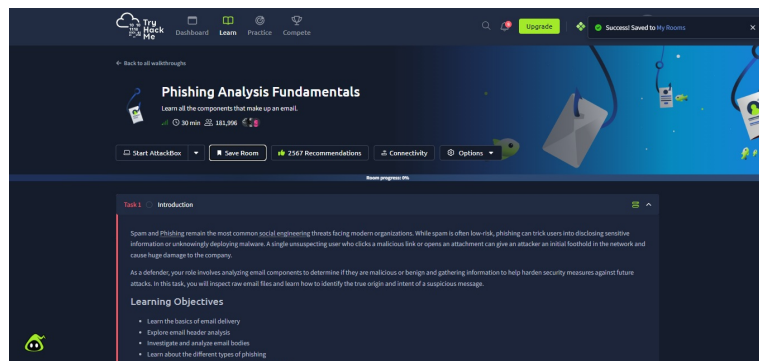
6. TryHackMe Room Completion Proof

Room: **Phishing Analysis Fundamentals** — completed (free-tier room, as required).

- thm1.png — Room overview / start
- thm2.png — LinkedIn post confirming module completion

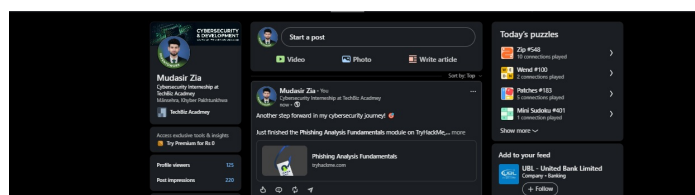
7. Screenshots

TryHackMe Room



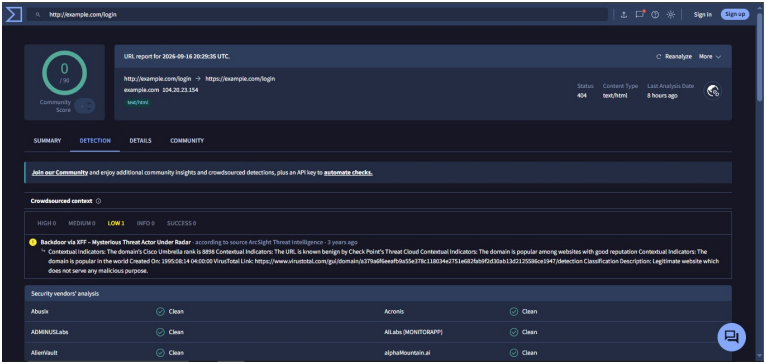
TryHackMe Room Overview

Completion Proof (LinkedIn)



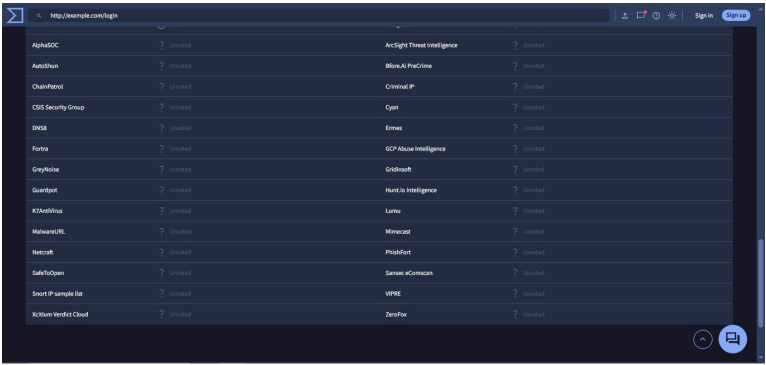
Completion Proof

VirusTotal URL Check — Detection Tab



VirusTotal Detection

VirusTotal URL Check — Vendor List



VirusTotal Vendors

End of report.